



of anti-aircraft fire a battleship or cruiser could throw up was daunting. But the vulnerability of surface ships to air attack, in the absence of adequate air cover, still came as a shock. One of the worst days of the war for the Royal Navy was on December 10, 1941, when the battleship *Prince of Wales* and the battle-cruiser *Repulse* were attacked by Japanese Mitsubishi G4M (Betty) medium bombers and Nakajima B5N (Kate) torpedo-bombers off the coast of Malaysia. The *Prince of Wales* alone had 175 anti-aircraft guns capable of firing 60,000 shells a minute, and the capital ships and their destroyer escorts were in open water, able to maneuver at speed. Yet in little over two hours, the *Prince of Wales* and the *Repulse* had been sunk, for the loss of only three Japanese aircraft.

Even Britain, which had lost its preeminence in carrier development, was able to score major victories by using aircraft against surface ships. The German battleship *Bismarck*, racing through the stormy North Atlantic for the safety of Brest Harbor in May 1941, would have eluded the pursuit of the Royal Navy had it not been spotted by a Catalina from RAF Coastal Command, and then damaged by Fairey Swordfish-delivered torpedoes from the carrier *Ark Royal*. That these slow-moving, open-cockpit biplanes were sent to attack the world's most high-tech battleship seems extraordinary in itself, let alone that they should have given it a crippling wound.

The Swordfish's other memorable success was the raid on the Italian fleet at Taranto in November 1940. Attacking a heavily defended shallow harbor by night, 21 of these seemingly obsolescent torpedo-bombers from the carrier *Illustrious* sank three battleships and a destroyer. A result out of all proportion to the force applied, the Taranto raid was studied with interest by naval experts around the world – including the Japanese.

STRIKING SWORDFISH

The strike on the Italian fleet at Taranto by Swordfish biplanes from the Royal Navy carrier Illustrious in November 1940 partly inspired the subsequent Japanese attack on Pearl Harbor.

Tora! Tora! Tora!

The surprise Japanese attack on Pearl Harbor on December 7, 1941, is one of the most celebrated, or infamous, uses of aircraft in the history of aviation. It was devised by senior Japanese commanders in a spirit of desperation, since Japan's determination to control China and Southeast Asia had put it on a collision course with the United States, a country that they could not realistically hope to defeat. Admiral Isoroku Yamamoto, the naval commander-in-chief, hoped that if he could take out the American Pacific fleet at the same time as grabbing Southeast Asia, Japan might at least buy time to organize a defense of its conquests. Admiral Nagumo, entrusted with commanding the surprise attack, was opposed to it and doubted that it would work.

Although Americans were understandably outraged at an attack timed to coincide with a declaration of war rather than follow it, the Pearl Harbor raid has to be recognized as a technically masterful naval air operation, in both its preparation and its execution. Japanese technical experts developed a torpedo that worked in the shallow water of the American harbor – normal torpedoes would have stuck in the seabed – and a bomb to pierce the battleships' armored decks, made by adding fins to an artillery shell. Their pilots, an elite group who had survived a harsh process of elimination, rehearsed the attack meticulously. And a fleet of 31 ships, including six aircraft carriers, was assembled and sailed undetected across 1,000 miles (1,600km) of ocean

JAPANESE AIRCRAFT USED IN THE RAID ON PEARL HARBOR

Aichi D3A ("Val")



The main dive-bomber type of the Imperial Japanese Navy at the time of Pearl Harbor, 126 "Vals" (as they were nicknamed by the Allies) took part in that attack. Aichi D3As sank more Allied naval ships than any other type of enemy aircraft during the war.

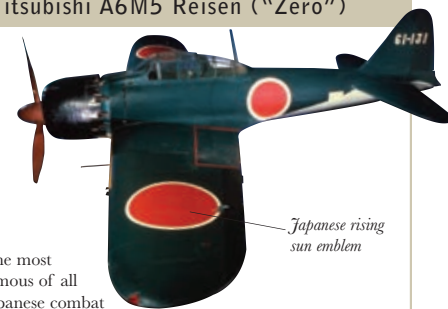
ENGINE 1,080hp Mitsubishi Kinsei 44 air-cooled 14 cylinder radial

WINGSPAN 47ft 1in (14.4m) **LENGTH** 33ft 5in (10.2m)

TOP SPEED 272mph (460kph) **CREW** 2

ARMAMENT 2 x 7.7-mm machine guns, 1 x 7.7-mm machine gun; 1 x 551-lb (250-kg) bomb, 2 x 66-lb (30-kg) bombs under wings

Mitsubishi A6M5 Reisen ("Zero")



The most famous of all Japanese combat aircraft, when the "Zero" first appeared in 1940, it outclassed every Allied fighter in the Pacific. This lightly built aircraft displayed outstanding maneuverability and had unparalleled range for the time.

ENGINE 1,300hp Nakajima NK1C Sakae 21 14-cylinder radial

WINGSPAN 36ft 1in (11m) **LENGTH** 29ft 11in (9.1m)

TOP SPEED 346mph (557kph) **CREW** 1

ARMAMENT 2 x 20mm cannon in wings, 2 x 7.7mm machine guns in fuselage; wing racks carry 2 x 132lb (60kg) bombs

Nakajima B5N2 ("Kate")

Whereas the British Navy's torpedo-bomber in 1940 was the antiquated Swordfish biplane, the Japanese developed this sleek all-metal monoplane design, bristling with modern features. After playing a major role at Pearl Harbor, it went on to help in the sinking of the aircraft carriers *Yorktown*, *Lexington*, and *Hornet*.



ENGINE 1,115hp Nakajima Sakae 21 air-cooled 14 cylinder radial

WINGSPAN 50ft 11in (15.5m) **LENGTH** 33ft 9in (10.3m)

TOP SPEED 235mph (378kph) **CREW** 3

ARMAMENT 2 x 7.7-mm machine guns, 1 x 7.7-mm machine gun in rear cockpit; 1 x 800-kg (1,764-lb) torpedo under centerline